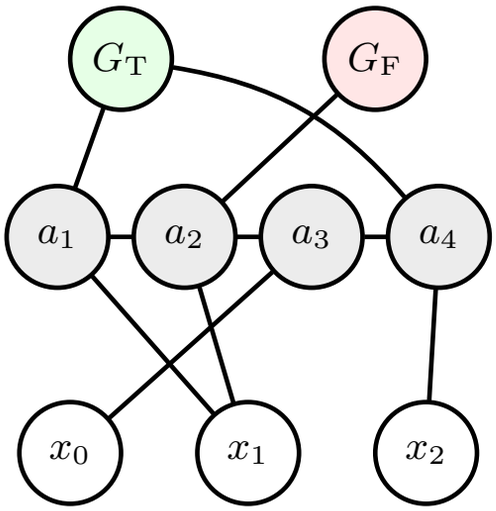
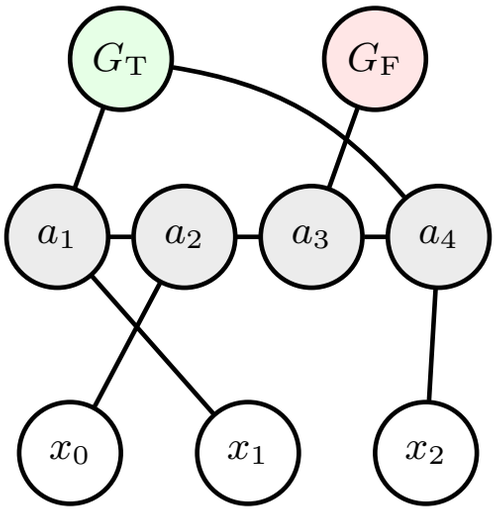
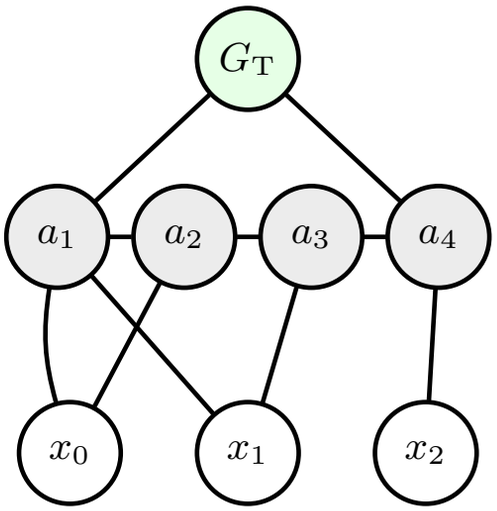


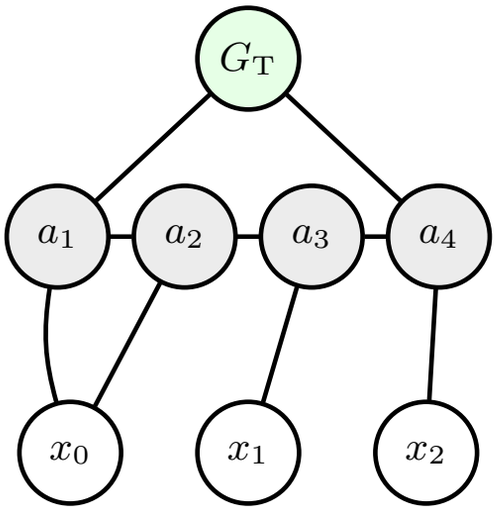
(a) 10 edges



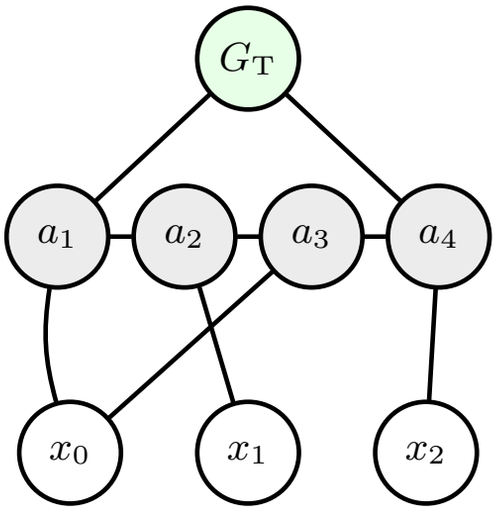
(b) 9 edges



(c) 10 edges



(d) 9 edges



(e) 9 edges

Valid clause gadgets for $C = (x_0 \vee x_1 \vee x_2)$ with four auxiliary vertices $a_1, \ldots, a_4$, one for each of the five classes up to relabeling the auxiliary vertices and the literals. In every class the auxiliary vertices form a path whose two ends are adjacent to G_{T}. None of them is S_3-symmetric, and no valid gadget has three or fewer auxiliary vertices. The palette triangle, the vertex G_{B}, and its edges to the literals are omitted; the edge counts exclude them.